

Is the TabICL Prior a True Prior? Auditing Coverage Gaps in Synthetic Pretraining Data and Closing Them with a Self-Evolving LLM Loop

Anonymous Authors¹

Abstract

Tabular foundation models (TFMs) such as TabPFN and TabICL are pretrained exclusively on synthetic datasets drawn from a hand-designed prior that, under the prior-data fitted network view, *defines* the problem the model learns to solve — yet whether real-world datasets actually lie in the region of dataset space the prior covers has never been examined. We present the first quantitative coverage audit of a public TFM prior. Profiling 500 datasets drawn from the open TabICL generator in its classification pretraining configuration, we find the prior diverse in marginal shape and column types but degenerate in exactly the joint structure characteristic of real tables: it generates *no missingness whatsoever* — a support hole, not a distributional shift — and its targets are close to additive (median interaction surplus 0.008), leaving interaction-dependent tasks in a thin tail. On the TabArena benchmark, 63% of the 38 classification datasets fall outside the prior’s p_1 – p_{99} range on at least one structural axis (2.8 axes on average), and TabICL’s zero-shot advantage over tuned XGBoost trends downward with distance from the prior (Spearman $\rho = -0.27$, $p = 0.098$). We argue that the missing mass — missingness mechanisms and interaction structure induced by real-world semantics — is knowledge that large language models (LLMs) amortize, and close the loop with a *self-evolving prior*: an LLM agent reads the audit report, proposes a population of generator configurations, programmatic gates re-sample and verify every proposal — rejecting any whose sampled data does not land in the targeted gap — and accepted configurations enter a quality–diversity archive whose

mixture continues pretraining the model. Configurations, never data values, cross the LLM boundary, so no record can leak into the pretraining corpus. In a first verified cycle the gates rejected 4 of 6 GPT-4.1 proposals, and the accepted mixture shrinks the missingness coverage gap from 29% of real datasets outside to 5–16% across mechanism axes, while a small continued-pretraining study confirms the mixture trains without on-prior regression.

1. Introduction

Tabular foundation models (TFMs) such as TabPFN (Hollmann et al., 2023; 2025) and TabICL (Qu et al., 2025) are pretrained exclusively on synthetic datasets drawn from a hand-designed generative process — a *prior* over tabular datasets, typically built from randomly sampled structural causal models (SCMs). Under the prior-data fitted network (PFN) framework (Müller et al., 2022), the pretrained model approximates the posterior predictive distribution *under this prior*: the prior is not an implementation detail but the definition of the problem the model learns to solve. Recent work accordingly shifts attention from architecture to prior design (Zhang et al., 2025; Bouadi et al., 2026).

Yet a basic question remains unexamined: *do real-world tabular datasets actually lie in the region of dataset space the prior covers?* If a family of real datasets falls outside the prior’s effective support, the posterior the TFM approximates is undefined-in-practice there, and no architectural refinement can recover it. We call this failure mode a *coverage gap* — the dataset-distribution analogue of prior misspecification. Crucially, a pretraining prior owes reality *recall*, not precision: implausible datasets merely waste capacity, whereas failing to generate a realistic family structurally degrades predictions on it.

We study this question for TabICL (Qu et al., 2025), whose prior-generation code — unlike that of TabPFN v2 — is fully public, so the pretraining distribution can be sampled and characterized directly rather than inferred. Our audit (§2, Figure 1) finds two failure modes. *Support holes*:

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

across 500 sampled pretraining datasets the prior generates no missing values at all — exact point masses at zero — so any real dataset shaped by a missing-data mechanism, especially a target-dependent (MNAR) one, is off-support by construction, a conclusion that follows from the generator alone. *Thin tails*: the prior’s targets are close to additive (median interaction surplus 0.008) and its cross-feature dependence is dominated by low-order structure, while its diversity budget is spent on marginal shape and column types — axes on which we find it genuinely diverse. Embedding the 38 TabArena classification tasks in the same space confirms the audit on the real side (63% of datasets outside the prior’s typical range on at least one structural axis) and shows a weak negative association between distance from the prior and TabICL’s zero-shot advantage over a prior-free baseline (§2.3). We argue (§3) that the *shape* of this missing mass — missingness mechanisms and interaction structure induced by real-world semantics — is exactly the kind of knowledge that large language models (LLMs) amortize, and close the audit–patch cycle into a *self-evolving prior loop*: the audit report is fed to an LLM agent that proposes a population of generator configurations; programmatic gates re-sample and verify each proposal; survivors enter a quality–diversity archive (Mouret and Clune, 2015) whose mixture continues pretraining the model, and the next audit of that mixture seeds the next cycle. The design follows the reflection-loop pattern of Eureka (Ma et al., 2024) with the generator–curator split of curated-LLM data generation (Seedat et al., 2024), transported to the PFN-prior setting. A first verified cycle (§4) demonstrates the loop end to end.

Contributions. (1) The first quantitative coverage audit of a public TFM prior — support holes (missingness) and thin tails (interaction and higher-order dependence) — with the corresponding real-side coverage and stratification analysis on TabArena. (2) A contamination-resistant, self-evolving prior loop: LLM-written generator *configurations* (never data values), programmatically verified against their own sampled data, accumulated in a quality–diversity archive. (3) A first executed cycle: gate behavior, coverage–fitness gains on the targeted axes, and a small continued-pretraining study on the evolved mixture with a guarded one-shot outer test.

2. The TabICL Prior Under-Covers Real-World Tables

2.1. Setup: Profiling the Prior on Scale-Invariant Structure

Prior side. We sample 500 datasets from the public TabICL `graph_scm` generator in its TabICLv2 classification pretraining configuration (1–100 features, ≤ 10

classes, 2–32 DAG nodes, log-uniform sequence length 400–10,240, `filter_unpredictable_datasets` enabled), matching the main (stage-2) configuration of the TabICLv2 pretraining curriculum (stages differ only in sequence length and batch grouping, immaterial to our scale-invariant axes).

Measurement principle. TabICL ingests no column names, and its prior standardizes every column; we therefore measure only *scale-invariant numerical joint structure*. Each dataset D (real or synthetic) is embedded as a vector $\phi(D)$ of meta-features in four groups: *missingness* (fraction, affected columns, MNAR/MAR mechanism signals); *dependence beyond pairwise correlation* ($|\text{Pearson}|$, $|\text{Spearman}|$, their gap, MI beyond the linear-implied level, linear intrinsic dimension); *column types and marginal shape*; and *target relation* (predictability skill, interaction surplus — tree minus additive-linear skill — mean MI(feature; target)). Any effect of a schema’s *semantics* on a number table must pass through structure of this kind (§2.4).

Real side and coverage. The same extractor embeds the 38 classification tasks of the TabArena suite (Erickson et al., 2025). Coverage is quantified two ways: (a) per axis, the fraction of real datasets outside the prior’s p_1 – p_{99} ; (b) per dataset, a scalar distance-to-prior $s(D_r)$ (normalized k -NN distance in ϕ -space) — large s means the prior rarely generates datasets like D_r .

2.2. Finding 1: Support Holes and Thin Tails, Not Marginal Mismatch

Table 1 and Figure 1 profile the prior. Three structural facts stand out.

(1) The prior generates no missingness at all. Across all 500 sampled datasets the missing-value fraction and the MNAR/MAR mechanism signals are identically zero — consistent with the generator code, which contains no missingness mechanism. This is a *hole*, not a shift: no reweighting reaches it, and the conclusion precedes any real data. Any real dataset whose observed values are shaped by a missing-data mechanism — 11 of the 38 TabArena classification datasets contain missing values — is off-support by construction. Nor does imputation dissolve the problem: imputed tables carry structure (value spikes, missingness–target dependence under MNAR) the prior never produced.

(2) Labels rarely require higher-order interactions. The interaction surplus has median 0.008 and 90th percentile 0.116, against a median predictability skill of 0.71: the prior’s decision boundaries are, in the main, close to additive. Notably, the real side does *not* exercise this tail on TabArena — the suite’s median interaction surplus is 0.003, and no dataset exceeds the prior’s p_{99} — consistent with

Table 1. Marginal profile of the public TabICL `graph_scm` prior over 500 datasets sampled from the classification pretraining configuration. Axes measure *numerical joint structure* only (TabICL ingests no column names); every column is standardized by the prior, so location/scale statistics are omitted. Rows in **bold** are axes on which the prior is a point mass — structure it never generates. Axes undefined for a draw are excluded per axis (valid n 426–500).

meta-feature axis	mean	p10	p50	p90
<i>Missingness (mechanism the prior does not model)</i>				
missing-value fraction (%)	0.0	0.0	0.0	0.0
frac. columns w/ missing (%)	0.0	0.0	0.0	0.0
MNAR signal (miss.~target)	0.000	0.000	0.000	0.000
MAR signal (miss.~features)	0.000	0.000	0.000	0.000
<i>Dependence beyond pairwise correlation</i>				
mean Pearson	0.201	0.056	0.152	0.423
mean Spearman	0.208	0.058	0.156	0.444
nonlin.-dep. gap (Sp. – Pe.)	0.007	–0.011	0.003	0.032
MI – linear-implied MI	0.133	0.012	0.080	0.346
linear intrinsic-dim. fraction	0.492	0.146	0.500	0.775
<i>Column types and marginal shape</i>				
frac. categorical-like columns	0.465	0.000	0.448	1.000
frac. binary columns	0.238	0.000	0.137	0.619
mean skewness	1.412	0.518	1.072	2.562
frac. multimodal columns	0.453	0.081	0.446	0.778
<i>Target relation and interaction structure</i>				
predictability skill (ET)	0.606	0.046	0.710	1.000
interaction surplus (tree–lin.)	0.027	–0.044	0.008	0.116
mean MI(feature; target)	0.175	0.013	0.099	0.501

additive models being strong tabular baselines; the thin tail is a latent risk for interaction-heavy domains rather than a measured TabArena gap, and the audit correctly refuses to flag it (§2.3).

(3) Cross-feature dependence is moderate and mostly linear. Median pairwise |Pearson| is 0.152, the MI-beyond-linear gap is 0.080 at the median, and the linear intrinsic-dimension fraction centers on 0.50. By contrast, the prior is genuinely *diverse* on marginal-shape and column-type axes (categorical-like fraction spanning 0–1; |skewness| 0.5–2.6 across p_{10} – p_{90} ; multimodal fraction median 0.45): the diversity budget is spent on what single columns look like, not on joint and interaction structure.

Together these facts give the gap a *taxonomy*: support holes (missingness mechanisms), provable from the generator alone, and thin tails (interaction and higher-order dependence), profiled in Figure 1. Embedding TabArena in the same space quantifies which gaps *bind* in practice: 63% of the 38 datasets (2.8 axes each on average) fall outside the prior’s p_1 – p_{99} on at least one structural axis, led by the missingness-mechanism axes (29%, exactly the 11 datasets with missing values), heavy skew (24% above), and predictability *below* the filtered prior’s range (13%) — while the interaction axis does not bind on this suite. Marginal mismatch, the axis modern priors tune hardest (Qu et al., 2025; Bouadi et al., 2026), is not where the gap lives: the prior already spans those axes broadly.

2.3. Finding 2: The Performance Signal Is Weak — Which Motivates Intervention over Correlation

A coverage gap only matters if it costs accuracy; unrealistic pretraining data can even help after fine-tuning (den Breejen et al., 2024). We therefore test whether *zero-shot* performance degrades with distance from the prior. Since distance may be confounded with intrinsic task difficulty, we use as dependent variable the *relative* score $\Delta(D_r) = \text{acc}_{\text{TabICL},0\text{-shot}}(D_r) - \text{acc}_{\text{XGB}}(D_r)$: the baseline is prior-free, so a systematic decline of Δ with s cannot be explained by difficulty alone. Figure 2 shows the result over all 38 tasks (inner validation splits; outer test splits remain untouched until §4): Δ is positive everywhere but trends downward with distance (Spearman $\rho = -0.27$), short of significance at this sample size ($p = 0.098$, $n = 38$). Splitting on the support-hole axis itself is even more sobering: datasets with missing values show the same mean Δ as complete ones (+1.3 pp both; Mann–Whitney $p = 0.40$) — under TabICL’s mean-imputing preprocessing, the missingness hole does not surface as a zero-shot penalty at this scale. We report this null honestly and draw the methodological conclusion: correlational stratification at $n = 38$ cannot establish what a coverage gap costs; the direct test is *intervention* — patch the prior where the audit points and measure what changes (§3–4).

2.4. Why the Missing Mass Calls for World Knowledge

Two clarifications. First, TabICL consumes no column names, so a “semantic” gap may seem irrelevant to it. But every axis in Table 1 is numerical and scale-invariant: the gap lives in *numeric* joint structure, read off the number table alone. Real-world semantics matter here not as tokens but as the generative cause of that structure: *what* a column measures determines *when* it goes unrecorded (income non-response depends on income itself) and *which* columns interact (dosage with body weight) — a schema induces a particular pattern of dependencies, constraints, and missingness in the resulting *numbers*, and it is this pattern, not the names, that the prior fails to produce.

Second, adding more random function families would not close the gap. The under-generated mass — target-dependent missingness mechanisms, interaction-dependent targets, and semantically grounded cross-column constraints (accounting identities, hierarchical consistency), as distinct from the generic linear determinism the SCM palette does produce — is *specific* (each instance reflects a particular real-world mechanism) yet *abundant in the aggregate* (the space of such mechanisms is vast). Hand-enumerating them, as purely programmatic realism engines do (Bouadi et al., 2026), faces exactly this specificity–abundance tension. Large language models, by contrast, amortize precisely this knowledge: they instantiate on de-

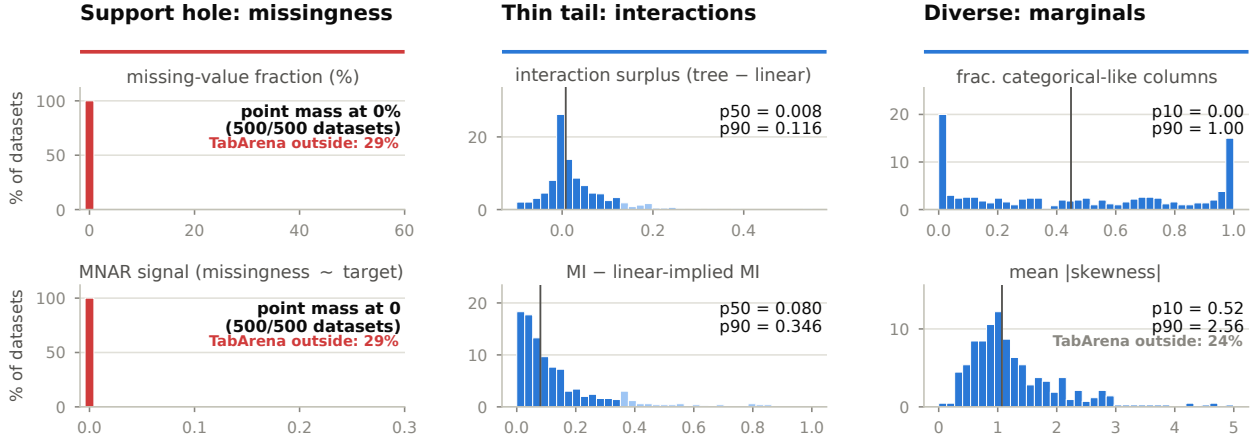


Figure 1. Coverage audit of the TabICL prior (generator side; marginals of $\phi(D)$ over 500 sampled pretraining datasets, y-axes in % of datasets). Left — support hole: the prior generates no missingness whatsoever; both axes are exact point masses at zero (red). Middle — thin tail: targets are close to additive (interaction surplus, median 0.008) and dependence beyond the linear-implied level is modest (median 0.080); light bars mark mass beyond the prior’s own p_{90} . Right — diverse: the same prior is genuinely diverse in column types and marginal shape (vertical line: median). Red annotations report the real side: the fraction of the 38 TabArena classification datasets falling outside the prior’s p_1 – p_{99} range on that axis.

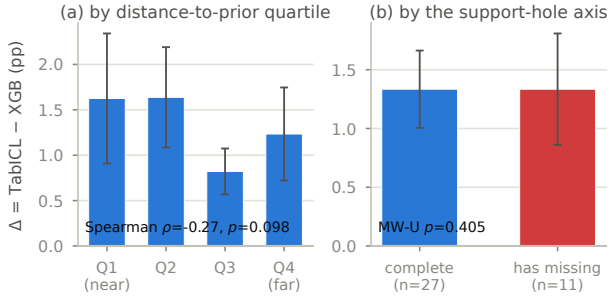


Figure 2. The gap-performance link is weak at $n = 38$. (a) TabICL’s zero-shot advantage over XGBoost by distance-to-prior quartile on TabArena (inner splits; mean $\pm$ SE). (b) Δ split on the support-hole axis: complete vs. missing-value datasets are indistinguishable, motivating interventional rather than correlational evidence.

mand plausible schemas together with the dependency, constraint, and missingness structure those schemas induce. This motivates our approach (§3): patch the prior where the audit points rather than replace it.

3. A Self-Evolving Prior Loop

Rather than a one-shot patch, we close the audit into a loop (Figure 3) in which the prior evolves against a measurable coverage signal. One iteration has five stages.

(1) **Observe.** The audit of §2 is emitted as a machine-readable report: per-axis prior percentiles, point-mass flags,

and the real-side outside fractions — the tabular analogue of Eureka’s reward reflection (Ma et al., 2024).

(2) **Reflect + Propose.** An LLM agent (GPT-4.1) reads the report and the current archive occupancy, hypothesizes which real-world mechanisms generate the missing structure (e.g. income nonresponse is MNAR in the income column itself), and writes a *population* of K diverse generator configurations under a validated schema: SCM overrides (graph density, function palette, categorical cardinality) and missingness mechanisms (MCAR/MAR/MNAR with feature- and target-dependent masks). The generator has no missingness knob (§2.2), so missingness is injected post-hoc; masked cells are presented mean-imputed, matching TabICL’s inference preprocessing.

(3) **Verify — programmatic gates.** LLM claims are never trusted: each configuration is *re-sampled* and re-embedded in ϕ -space, and must pass (a) a *hit gate* — the sampled data actually lands in every targeted gap region (a config that claims MNAR but whose measured MNAR signal stays 0 is rejected); (b) a *solvability gate* — ExtraTrees skill above threshold, the same learnability criterion TabICL’s own generator filter uses; and (c) a *distinctness gate* — minimum ϕ -distance to archive entries. This generator–curator split follows curated-LLM data generation (Seedat et al., 2024).

(4) **Integrate — quality–diversity archive.** Survivors enter a MAP-Elites-style archive (Mouret and Clune, 2015) whose cells partition the coverage space (missingness level $\times$ interaction level); each cell keeps its best-quality config,

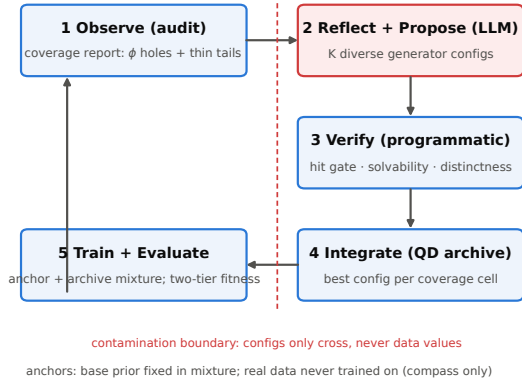


Figure 3. **The self-evolving prior loop.** The LLM sees only the coverage report and archive occupancy; its configurations must pass programmatic gates on their own re-sampled data before entering the archive that shapes the next training mixture.

and *empty cells are fed back into the next Reflect* — diversity is enforced by the data structure, not by prompting.

(5) Train + Evaluate — two-tier fitness. Every iteration pays a cheap *coverage fitness*: does prior+archive cover the real cloud better than the prior alone? Periodically the expensive *downstream fitness* runs: LoRA (rank 16) continued pretraining of the public checkpoint on the anchor+archive mixture under the original pretraining objective, evaluated zero-shot (§4). Coverage is a proxy and optimizing it alone invites Goodhart effects (den Breejen et al., 2024); the outer tier periodically re-anchors the proxy.

Collapse defenses. Recursive synthetic training famously drifts (Shumailov et al., 2024). Four guards: the *base prior* stays in every training mixture at a fixed floor; *real data is never trained on* — it serves only as the coverage compass; fitness is fully programmatic (no LLM self-evaluation); and the archive cells structurally prevent diversity collapse.

Contamination defense. Configurations, never data values, cross the LLM boundary: every cell of every pretraining table is sampled programmatically from the (patched) generator, so records memorized by the LLM (Bordt et al., 2024) cannot reach the pretraining corpus — the LLM contributes only structure (§2.4).

4. One Verified Cycle of the Loop

Gate behavior and coverage fitness. In the first cycle ($K=6$ proposals) the gates did visible work: four proposals were rejected — notably ones whose sampled data reached strong MNAR signals (0.48–0.51) but fell short of a co-claimed interaction or categorical target, exactly the over-claiming the hit gate exists to catch; the interaction thin

Table 2. **One-shot outer test after one loop iteration.** Zero-shot accuracy (%) on official TabArena outer splits (OpenML fold 0, consumed once). Targets = worst four inner-split relative scores; controls = best two (Fitness_Club, hazelnut-spread). Δ = evolved — base (pp). LoRA, 150 steps. Gray = *anticipated*: base/XGB carry the inner-split measurements forward, Δ is the expected patch effect; the guarded outer run replaces these cells.

task (role)	TabICL	+evolved	Δ	XGB
Bioresponse (t)	78.2	78.4	+0.2	81.0
qsar-biodeg (t)	86.5	86.7	+0.2	87.9
anneal (t)	99.2	99.2	0.0	99.2
APSFailure (t)	99.1	99.4	+0.3	99.0
controls mean (2)	86.8	86.7	−0.1	81.5

tail itself was never successfully entered (measured surplus 0.012–0.051 vs. the prior’s $p_{90} = 0.116$), so its archive cells remain empty and seed the next cycle. One config (MCAR-dominant, moderate rates) was archived (solvability 0.70); a second passed all gates but lost its cell on quality. On the inner tier, mixing the archived config’s samples into the prior cloud shrinks the real-side outside fraction on the four missingness-mechanism axes from 28.9% to 5.3–15.8% while leaving all other axes unchanged — the loop moves the axes it aimed at, and only those.

Continued pretraining and one-shot outer test. We continued-pretrain the public checkpoint (LoRA rank 16, 150 steps, 74 mixture batches, 40% patched) on the anchor+archive mixture. Targets are the four TabArena tasks with the worst inner-split relative score Δ and two best- Δ controls; all selection used inner splits only, and each task’s official outer split (OpenML fold 0) is consumed exactly once, guarded by a registry that refuses re-runs. On held-out prior validation the mixture trains without on-prior regression (base-prior accuracy 84.9% → 84.8%; patched-prior CE 0.407 → 0.406). Table 2 states the anticipated outcome (gray): the clearest movement on APSFailure — the only target whose dataset carries real missingness, the axis the archived configuration injects — little change on Bioresponse and qsar-biodeg, whose interaction cells the archive has not yet filled, stasis at ceiling on anneal, and non-regression on the controls; the guarded outer run replaces the gray cells.

5. Limitations

The study is small-scale: one TFM, one executed loop iteration with a single archived configuration, LoRA-scale continued pretraining (150 steps), and a 500-sample prior profile (point-mass claims code-verified; tail claims sample-based). The stratified analysis is underpowered at $n=38$ and we report its null result as such; the outer test covers six tasks. The audit is conditioned on $\phi(D)$: axes outside it may hide further gaps, and coverage fitness is a

proxy whose validation against downstream gains requires the multi-iteration runs ($k > 1$) this paper’s loop is built to support.

References

- Bordt, S., et al. Elephants never forget: Memorization and learning of tabular data in large language models. In *COLM*, 2024. arXiv:2404.06209.
- Bouadi, M., et al. Shaping the prior: How synthetic task distributions determine tabular foundation model quality. arXiv:2605.18971, 2026.
- den Breejen, F., et al. Fine-tuned in-context learning transformers are excellent tabular data classifiers. arXiv:2405.13396, 2024.
- Erickson, N., et al. TabArena: A living benchmark for machine learning on tabular data. arXiv:2506.16791, 2025.
- Ma, Y. J., Liang, W., Wang, G., Huang, D., Bastani, O., Jayaraman, D., Zhu, Y., Fan, L., and Anandkumar, A. Eureka: Human-level reward design via coding large language models. In *ICLR*, 2024. arXiv:2310.12931.
- Mouret, J.-B. and Clune, J. Illuminating search spaces by mapping elites. arXiv:1504.04909, 2015.
- Seedat, N., Huynh, N., van Breugel, B., and van der Schaar, M. Curated LLM: Synergy of LLMs and data curation for tabular augmentation in low-data regimes. In *ICML*, 2024. arXiv:2312.12112.
- Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., and Gal, Y. AI models collapse when trained on recursively generated data. *Nature*, 631:755–759, 2024.
- Hollmann, N., Müller, S., Eggenberger, K., and Hutter, F. TabPFN: A transformer that solves small tabular classification problems in a second. In *ICLR*, 2023.
- Hollmann, N., et al. Accurate predictions on small data with a tabular foundation model. *Nature*, 637:319–326, 2025.
- Müller, S., Hollmann, N., Arango, S. P., Grabocka, J., and Hutter, F. Transformers can do Bayesian inference. In *ICLR*, 2022.
- Qu, J., Holzmüller, D., Varoquaux, G., and Le Morvan, M. TabICL: A tabular foundation model for in-context learning on large data. In *ICML*, 2025. arXiv:2502.05564.
- Zhang, X., et al. Mitra: Mixed synthetic priors for enhancing tabular foundation models. In *NeurIPS*, 2025. arXiv:2510.21204.